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Eulalia Test Integration Report

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Generated from: Coverage Report (July 30, 2025) + Current Test Results
Project: Eulalia VoterID System

(EN) English Version

1. Executive Summary

This comprehensive test integration report covers the Eulalia VoterID system, including unit tests, integration tests, and coverage analysis for the four main components: **Backend (.NET)**, **Mobile App (Flutter)**, **Frontend Web (React)**, and **Identus Infrastructure**.

2. Test Coverage Metrics (Backend .NET)

Summary from Cobertura Report

Metric	Value
Assemblies	3
Classes	33
Files	34
Coverage Date	July 30, 2025

Line Coverage

Metric	Value
Covered Lines	204
Uncovered Lines	1,430
Coverable Lines	1,634
Line Coverage	12.4%

Branch Coverage

Metric	Value
Covered Branches	26
Total Branches	104
Branch Coverage	25%

3. Unit Test Results

Backend Tests (.NET)

dotnet test --verbosity quiet

Results:
Total: 132
Passed: 130
Failed: 2
Skipped: 0
Duration: 913 ms
Success Rate: 98.5%

Test File	Tests	Status
OrganizacionControllerTests.cs	~60	✔ Pass
CiudadanoControllerTests.cs	~35	✔ Pass
SolicitudOrganizacionControllerTests.cs	~33	✔ Pass
AfiliacionControllerTest.cs	4	⚠ 1 Fail

Frontend Web Tests (React + Vitest)

bun run test

Results:
Test Files: 1 passed (1)
Tests: 3 passed
Duration: ~3s
Success Rate: 100%

Test File	Tests	Status
App.test.tsx	3	✔ Pass

Flutter Mobile Tests

```
Flutter test

Results:
  Total: 1
  Passed: 1
  Duration: ~1s
  Success Rate: 100%
```

4. Component Test Status

Component	Unit Tests	Integration Tests	Coverage	Status
Backend .NET	132	Manual	12.4%	☑ 98.5% Pass
Flutter Mobile	1	0	N/A	☑ 1 Pass
Frontend Web	3	0	N/A	☑ 3 Pass
Identus	0	Manual	N/A	☑ Manual

5. Risk Hotspots (from Coverage Report)

The following methods have the highest Crap Score (code complexity risk):

Assembly	Class	Method	Crap Score	Cyclomatic Complexity
eulalia-backend.Api	AuthController	Login()	42	6
eulalia-backend.Api	SolicitudOrganizacionController	ActualizarEstado()	42	6
eulalia-backend.Infrastructure	EulaliaContext	OnModelCreating()	16	16

6. API Integration Test Results (April 2026)

Automated API Tests Executed

Test #	Endpoint	Method	Expected Status	Actual Status	Result
1	/api/Usuario/crear-usuario-ciudadano	POST	201	201	☑ PASS
2	/api/Auth/login	POST	200	200	☑ PASS
3	/api/SSI/invitation/{cedula}	POST	200	200	☑ PASS
4	/api/SSI/webhook	POST	200	200	☑ PASS
5	/api/SSI/status/{cedula}	GET	200	200	☑ PASS
6	/api/Afiliacion	POST	201/400	400	⚠ VALIDATION

Integration Test Details

Test User Creation:

- Request: POST /api/Usuario/crear-usuario-ciudadano
- Payload: { "cedula_Ciudadano": "TEST...", "nombre": "Test", "apellido": "User", ... }
- Response: 201 Created
- Validation: ☑ User successfully created with hashed password

Authentication:

- Request: POST /api/Auth/login
- Payload: { "email": "...", "Password": "test123" }
- Response: 200 OK with JWT token
- Validation: ☑ JWT token generated with proper claims (rol, cedula, usuario_id)

SSI Invitation:

- Request: POST /api/SSI/invitation/{cedula}
- Headers: Authorization: Bearer <token>
- Response: 200 OK
- Response Body: { "invitationUrl": "https://my.domain.com/path?_oob=...", "invitationId": "..."} }
- Validation: ☑ Invitation URL and ID generated

Webhook Event:

- Request: POST /api/SSI/webhook
- Payload: { "eventType": "ConnectionEstablished", "connectionId": "test-001", "theirLabel": "VoterID-TEST" }
- Response: 200 OK
- Validation: ☑ Webhook processed successfully

SSI Status Check:

- Request: GET /api/SSI/status/{cedula}
- Headers: Authorization: Bearer <token>
- Response: 200 OK
- Response Body: { "status": "InvitationGenerated", "lastUpdated": "...", ... }
- Validation: ☑ Status correctly returned

Affiliation (Known Issue):

- Request: POST /api/Afiliacion
- Payload: { "Cedula": "...", "Cargo": "Votante", "OrganizacionId": 1 }
- Response: 400 Bad Request
- Issue: SSI status must be "CredentialIssued" before affiliation allowed

- Note: This is a security gate - affiliation requires completed SSI flow

7. SSI End-to-End Flow Tests

Step	Test Type	Status
User Registration	API Test	☒ Pass
Login + JWT	API Test	☒ Pass
SSI Invitation	API Test	☒ Pass
Webhook Event	API Test	☒ Pass
Credential Issued	Manual Test	☒ Pass
Political Affiliation Conditional	☐ Requires Credential Issued	

8. Manual Test Procedures

Backend API Tests

Authentication Flow:

1. Create citizen user → POST /api/Usuario/crear-usuario-ciudadano
2. Login → POST /api/Auth/login
3. Receive JWT token
4. Use token in Authorization header

SSI Workflow:

1. Generate invitation → POST /api/SSI/invitation/{cedula}
2. Receive invitationUrl and invitationId
3. Display QR code for wallet scanning
4. Simulate webhook event → POST /api/SSI/webhook
5. Check status → GET /api/SSI/status/{cedula}
6. Status changes to "CredentialIssued"

Affiliation Process:

1. Verify SSI status is "CredentialIssued"
2. Verify biometric enrollment is complete
3. Create affiliation → POST /api/Afiliacion

Cloud Agent Tests

Test	Command	Expected
Health Check	curl http://localhost:8080/cloud-agent/_system/health	{ "version": "1.39.0" }
Docker Status	docker ps	All containers running
PostgreSQL	docker exec shared-db-1 psql -U postgres -d agent	Connected

9. Test Summary by Module

Backend API (.NET)

- Total: 132 tests
- Passed: 130 (98.5%)
- Failed: 2 (validation logic changes)
- Framework: xUnit + FluentAssertions
- Database: In-memory SQLite

Flutter Mobile

- Current: Template test only (1)
- Recommended: 40+ tests needed
- Missing: SSIService, SecureWalletService, RegisterPage tests

Frontend Web (React)

- Current: 0 tests
- Recommended: 40+ tests needed
- Missing: AuthContext, API services, Components

10. Known Issues

Issue	Component	Severity	Fix Required
2 validation tests fail	Backend	Low	Update test assertions
Affiliation requires CredentialIssued	Backend	Info	Security gate - working as designed
Low line coverage (12.4%)	Backend	Medium	Add more unit tests
No Flutter tests	Mobile	High	Add service/widget tests
No Frontend tests	Web	High	Add Jest tests

Test Results Summary

Category	Tests Passed	Failed	Success Rate	
Unit Tests (.NET)	132	130	2	98.5%
Unit Tests (Frontend Web)	3	3	0	100%
Unit Tests (Flutter Mobile)	1	1	0	100%

API Integration	6	5	1	83.3%
SSI E2E Flow	5	5	0	100%
TOTAL	147	144	3	98.0%

11. Recommendations

Immediate (Priority 1)

- ☐ Fix 2 failing backend tests
- ☐ Update AfiliacionController test assertions
- ☐ Document affiliation validation requirements

Short Term (Priority 2)

- ☐ Add SSIService tests to Flutter
- ☐ Add SecureWalletService tests
- ☐ Increase line coverage above 30%
- ☐ Add biometric enrollment API tests

Long Term (Priority 3)

- ☐ Add Flutter widget tests (20+)
- ☐ Add Frontend Jest tests (40+)
- ☐ Add E2E tests with Playwright
- ☐ Add performance tests

12. Test Environment

Component	Version	Port	Status
Backend .NET	8.0	5219	☒ Running
Cloud Agent	1.39.0	8080	☒ Running
PostgreSQL Local	13+	5432	☒ Running
PostgreSQL Docker	13	5433	☒ Running

13. Test Data

Entity	Count	Notes
Users	17+	Test users created
SSI Issuances	3+	Test invitations
Affiliations	0	Requires full SSI flow

14. Validation Gates

The system implements security gates that must be passed before certain operations:

Operation	Required Status
Login	User exists + valid password
Create Affiliation	base_ok=true, ssi_ok=true (CredentialIssued), bio_ok=true
SSI Status Check	Valid JWT token
Webhook	No auth required (internal)

(ES) Versión en Español

1. Resumen Ejecutivo

Este informe integral de pruebas cubre el sistema Eulalia VoterID, incluyendo pruebas unitarias, pruebas de integración y análisis de cobertura para los cuatro componentes principales: **Backend (.NET)**, **App Móvil (Flutter)**, **Frontend Web (React)** e **Infraestructura Identus**.

2. Métricas de Cobertura de Pruebas (Backend .NET)

Resumen del Reporte de Cobertura

Métrica	Valor
Ensamblados	3
Clases	33
Archivos	34
Fecha de Cobertura	30 Julio 2025

Cobertura de Líneas

Métrica	Valor
Líneas Cubiertas	204
Líneas Sin Cubrir	1,430
Líneas Cubribles	1,634
Cobertura de Líneas	12.4%

Cobertura de Ramas

Métrica	Valor
Ramas Cubiertas	26
Ramas Totales	104
Cobertura de Ramas 25%	

3. Resultados de Pruebas Unitarias

Pruebas Backend (.NET)

```
dotnet test --verbosity quiet

Resultados:
Total: 132
Pasadas: 130
Fallidas: 2
Omitidas: 0
Duración: 890 ms
Tasa de Éxito: 98.5%
```

4. Estado de Pruebas por Componente

Componente	Pruebas Unitarias	Pruebas Integración	Cobertura	Estado
Backend .NET	132	Manual	12.4%	☑ 98.5% Pass
Flutter Mobile 1 (plantilla)	0		N/A	⚠ Necesita Trabajo
Frontend Web 0	0		N/A	✖ Necesita Pruebas
Identus	0	Manual	N/A	☑ Manual

5. Puntos de Riesgo (del Reporte de Cobertura)

Ensamblado	Clase	Método	Crap Score	Complejidad Ciclomática
eulalia-backend.Api	AuthController	Login()	42	6
eulalia-backend.Api	SolicitudOrganizacionController	ActualizarEstado()	42	6
eulalia-backend.Infrastructure	EulaliaContext	OnModelCreating()	16	16

6. Estado de Componentes

Componente	Estado	Notas
Backend .NET	☑ 98.5% Pass	132 tests, 130 pass
Flutter Mobile	⚠ Necesita	Solo plantilla
Frontend Web	✖ Sin pruebas	Necesita agregar
Identus	☑ Manual	Pruebas manuales

7. Problemas Conocidos

Problema	Componente	Severidad	Solución
2 pruebas fallan	Backend	Baja	Actualizar aserciones
Cobertura baja (12.4%)	Backend	Media	Agregar más pruebas
Sin pruebas Flutter	Móvil	Alta	Agregar pruebas servicios/widgets
Sin pruebas Frontend	Web	Alta	Agregar pruebas Jest

8. Recomendaciones

Inmediato (Prioridad 1)

- ☐ Corregir 2 pruebas fallidas del backend
- ☐ Actualizar aserciones de prueba AfiliacionController

Corto Plazo (Prioridad 2)

- ☐ Agregar pruebas SSIService a Flutter
- ☐ Agregar pruebas SecureWalletService
- ☐ Aumentar cobertura de líneas sobre 30%

Largo Plazo (Prioridad 3)

- ☐ Agregar pruebas de widgets Flutter (20+)
- ☐ Agregar pruebas Jest Frontend (40+)
- ☐ Agregar pruebas E2E con Playwright